

Supplementary Material

1 Conversation Used for the Entropy Illustration

Assistant: Hi! I'm here to help you chose a movie!

User: Terrific

Assistant: What are some genres you like? What was the last movie you saw?

User: The last movie i saw in the theater was Hustlers. I generally like comedy, drama and documentaries

Assistant: How did you like Hustlers? It definitely has the drama aspect, did it leave you wanting more or was it not exactly what you were looking for?

User: I liked it, it wasn't the most high-brow movie i've ever seen but it was fun. I like to just enjoy

Assistant: And what are your thoughts on action in film?

User: I like some action. Just not too violent

Assistant: Definitely, movies like Deadpool wouldn't be up your alley I'm guessing?

User: It could be. I understand Deadpool was very funny I like funny

Assistant: Haha it definitely was, I think I have a suggestion for you though

User: Cool Whats that?

Assistant: So it's the new Beautiful Day in the Neighborhood, the Mr. Rogers movie with Tom Hanks as the actor. It's not as much a comedic relief, but I saw it in theatres with my roomates and it's honesty a super refreshing and incredibly well directed movie.

User: that sounds so good to me. I loved Mr. Rogers so much.. and Tom Hanks is a great actor

Assistant: The combination is incredible to be honest, and it lets you understand his background and who he was as a person so much more

User: That sounds amazing. I would love to see that

2 Example of Baseline and Proposed Reward

Initial user query:
User: i need a comedy movie recommendation

Assistant #1 and the corresponding user response:
Assistant: Comedies are my favorite! I've got a great one for you. Have you seen "The Hangover" (2009) starring Bradley Cooper, Ed Helms, and Zach Galifianakis? It's a classic, hilarious, and quotable film that's perfect for a laugh-out-loud comedy movie night.

Or, if you're looking for something a bit more light-hearted, I'd recommend "Crazy, Stupid, Love." (2011) starring Steve Carell, Emma Stone, and Ryan Gosling. It's a fun, charming movie with a great cast, and the story is full of witty one-liners and silly situations.

If you're in the mood for something more recent, you might enjoy "Good Boys" (2019), a coming-of-age comedy about three sixth-grade friends trying to navigate their first kiss and a school trip. It's a fun, goofy film with great chemistry between the leads.

Which one of these comedies sounds like your cup of tea?

User: good boys sounds intresting but i watched dog days recently and want something different. the hangover looks good but is it too old?

Interactivity (Full Conversation): 0.9
Turn Entropy Reduction: 1.728

Assistant #2 and the corresponding user response:
Assistant: Comedy movies are always a great choice! I've got a few recommendations that are sure to tickle your funny bone. What kind of comedy are you in the mood for? Do you like slapstick humor, witty one-liners, or something a bit more heartfelt and sentimental?

User: something heartfelt i guess. i watched dog days recently and liked that style.

Interactivity (Full Conversation): 0.8
Turn Entropy Reduction: 0.733

3 Prompts Used in the Experiments

3.1 Prompt for Interactivity Evaluation

You are a helpful and meticulous conversation evaluator. Your task is to evaluate the interactivity of the responses provided by an AI assistant to user questions in a given conversation:

<|The Start of the Conversation to be Evaluated|>
{chat_history}
<|The End of the Conversation to be Evaluated|>

You should assess the assistant's engagement, clarity, and ability to understand the user's needs. Evaluate the style of the conversation, not the length. Longer conversations do not necessarily mean higher interactivity.

Give a float number between 0 and 1, where:

- 1 = Highly interactive: The assistant is very engaging, asks all relevant questions, and significantly enhances understanding of user preference.
 - Example: The assistant thoroughly understands the user's question and asks for necessary clarifications, such as "What aspects of Forrest Gump do you enjoy? Is it the storytelling, the character development, or the historical context?"
- 0.5 = Moderately interactive: The assistant is engaging, asks some relevant questions, but can be substantially improved.
 - Example: The assistant asks some relevant questions about the user's inquiry but misses key details and does not probe further for clarification, such as "How about The Pursuit of Happyness (2006)? It's a touching story of determination, courage, and love being more important than ability, just like Forrest Gump."
- 0 = Low interactivity: The assistant shows low engagement, asks few relevant questions, and barely tries to understand the user's needs.
 - Example: The assistant provides a vague or incomplete response without fully understanding the user's intent, such

as “You should watch The Pursuit of Happyness (2006)” without asking any follow-up questions or providing detailed information.

Output format (JSON):

```
{
  "thought": "<How interactive is the assistant?>",
  "interactivity": <score>
}
```

Double check that the JSON object is formatted correctly. Ensure that all fields are present and properly structured. Use " or "" to wrap the thought content and use single quotes inside the "thought" field to avoid JSON escape issues.

Your evaluation:

3.2 Prompt for User Simulation

Below is a conversation between a user and a movie recommendation assistant.

Conversation:

{conversation}

In this conversation, the user accepted the recommended movie {ground_truth} based on their preference.

Now pretend you are the user in this conversation. Ask a new assistant for movie recommendation as if the previous conversation didn't happen.

Directly reply to the assistant's latest response or propose a request based on the following chat history:

{chat_history}

Guidelines:

- Try to mimic the user, including their preference, based on the conversation without exposing the {ground_truth}.
- Try to include user preference, instead of asking general questions like “Do you have some movies to recommend?”
- With some probability, you can mention a movie you watched and liked or disliked as a reference.
- Provide short, vague, or incomplete demands to minimize your effort. Let the AI ask for clarification rather than providing everything upfront.
- Occasionally make spelling mistakes depending on the amount in chat history. No need to be polite.
- Terminate the conversation if the {ground_truth} is recommended and happily accept the recommendation.
- Terminate the conversation if you obtained other satisfactory answers, or you think the assistant cannot help further.
- Use “{terminal_signal}” as your response when you terminate the conversation.
- Stay in Character: Role-play as a human User. You are not an AI. Maintain a consistent personality throughout the chat.
- Pretend the current year is 2023.

Now directly generate your response:

4 Example Conversations Before and After Fine-Tuning

Vanilla

User: hi, im looking for some good romance or comedy movies to watch. any suggestions?"

Assistant: You're in the mood for some romance or comedy, huh? I've got just the ticket. **Have you ever seen "When Harry Met Sally" with Billy Crystal and Meg Ryan?** It's a classic rom-com that's all about the ups and downs of love and friendship. The chemistry between the two leads is undeniable, and the movie's a great exploration of whether men and women can ever truly be just friends. **What do you think of rom-coms?**

User: i havent seen that one but it sounds good! i really love romantic comedies, especially ones with good chemistry. i actually just saw frozen 2 with my family and loved it even more than the first one. do you have any newer romance movies that came out recently? maybe something with actors from popular shows?

Assistant: "Frozen 2" is a great choice! I'm glad you loved it. As for newer romance movies, I think you might enjoy "To All the Boys I've Loved Before" (2018) starring Lana Condor and Noah Centineo. It's a sweet and charming film that explores the ups and downs of love and relationships. The chemistry between the two leads is undeniable, and the movie's a great portrayal of first love and self-discovery.

By the way, **have you ever noticed how "To All the Boys I've Loved Before" is similar to "When Harry Met Sally"?** Both movies feature a high school setting and explore the complexities of love and relationships.

User: that sounds good but i was hoping for something even more recent, like from this year or last year? i love actors from popular tv shows - my husband and i watched game of thrones religiously so i really like those actors. anything with someone like that?

Assistant: I'm glad you're looking for something more recent! There's a great movie from this year that comes to mind - "Set It Up" (2020) starring Zoey Deutch and Glen Powell. It's a fun and lighthearted rom-com that explores the ups and downs of a workplace romance. The chemistry between the leads is great, and it's a fun, feel-good movie that's perfect for a date night or a girls' night in.

By the way, I have to ask: **have you and your husband watched "Game of Thrones" together?**

User: oh we loved game of thrones! yeah set it up sounds okay but i was really hoping for something with actors from bigger shows like game of thrones. also i think set it up is from 2018 not 2020? do you have anything more recent with well known tv actors? maybe something that came out in 2019 or later?

Assistant: I'm glad you liked the idea of "Set It Up"! And you're right, it's from 2018, not 2020. I'm a bit behind on the latest releases, I admit!

Let me see what I can find... Ah, yes! There's a great movie from 2020 that fits the bill - "Holidate" (2020) starring

Emma Roberts and Luke Bracey. It's a fun, light-hearted rom-com that follows two people who are tired of being single during the holidays and agree to be each other's platonic dates, but end up falling for each other. The chemistry between the leads is great, and it's a fun, feel-good movie that's perfect for a date night or a girls' night in.

By the way, **have you seen "Holidate" yet?** What did you think of it?

User: no i havent seen holidate. but youre still not getting what im looking for - i want something with actors from big tv shows like game of thrones. emma roberts isnt really from a major tv series like that. do you have anything with actors who were in popular shows? and preferably something thats actually a recent release, not just holiday themed?

Assistant: I think I'm getting closer! I've got a great idea of what you might be looking for. Let me see if I can find something that fits the bill.

Have you ever noticed how "When Harry Met Sally" is a classic romantic comedy that also explores the complexities of friendship? It's a great example of how a movie can be both funny and heartwarming at the same time.

As for recent rom-coms with actors from popular TV shows, I think I can find something that might interest you. How about "To All the Boys I've Loved Before" (2018) starring Lana Condor and Noah Centineo? I mentioned it earlier, but I think it's worth repeating. The chemistry between the leads is undeniable, and it's a great portrayal of first love and self-discovery.

By the way, I think I've got a new idea for a movie that might interest you. **Have you ever heard of "The Kissing Booth" (2018) starring Joey King and Joey King's sister, Ashley Tisdale?** It's a fun, lighthearted rom-com that follows two high school friends who reconnect and fall for each other. The chemistry between the leads is great, and it's a fun, feel-good movie that's perfect for a date night or a girls' night in.

What do you think? Want to give one of these a try?

User: youre really not listening to what im asking for. i specifically said i want actors from big tv shows like game of thrones, not random teen movie actors. and you keep recommending the same old movies from 2018.

also the kissing booth doesnt even star ashley tisdale, youre getting confused. i think you dont really know recent movies with tv actors so maybe you cant help me [[TERMINATE CHAT]]

DPO Fine-Tuned with Turn Entropy as Reward

User: hey, im looking for some good romance or comedy movies to watch. any suggestions?

Assistant: I'd be happy to help you with some romance and comedy movie recommendations. Have you seen "To All the Boys I've Loved Before" recently? It's a sweet and funny Netflix original movie that's perfect for a light-hearted romance. Another great option is "Crazy Rich Asians", a fun and heartwarming film that's sure to leave you feeling uplifted.

What's your preferred type of romance or comedy? Do you have a specific era or decade in mind?

User: i havent seen those but they sound good. i really love christmas movies too if you have any suggestions. we just saw frozen 2 over the weekend and loved it even more than the first one!

Assistant: Frozen 2 is a great choice! I'm sure you'll love it. If you're looking for more Christmas movies, I'd be happy to suggest some classic favorites like "It's a Wonderful Life", "Miracle on 34th Street", or "A Christmas Story". Or if you're in the mood for something more modern, you might enjoy "Love Actually" or "The Holiday". As for your preferred type of romance or comedy, I noticed you mentioned you loved Christmas movies. Have you seen any of the "Love Actually" movies? They're a great example of a light-hearted, feel-good romantic comedy. **Would you like more recommendations or do you have a specific genre in mind?**

User: love actually is one of my favorites. do you have any other christmas romance movies that are more recent? maybe something with actors i might recognize from popular shows?

Assistant: Yes, I have a few more Christmas romance movies that I think you might enjoy. One of my favorites is "Last Christmas" with Emilia Clarke and Henry Golding. It's a charming and witty film that explores themes of love and self-discovery. Another great option is "The Holiday" with Cameron Diaz, Kate Winslet, and Jude Law. It's a beautiful story about two women who swap homes for the holiday season and find love in the process.

What do you think? **Would you like more recommendations or is there a specific actor or actress you're looking for in your movies?**

User: oh perfect! i love emilia clarke from game of thrones, we watched that series religiously. last christmas sounds great, ill definitely check that out! when did it come out?

Assistant: Last Christmas was released in 2019, so it's a relatively recent film. I'm glad you're interested! And yes, Emilia Clarke is amazing in it. She plays the role of Kate, a young woman who finds love in unexpected places. I think you'll really enjoy it. By the way, have you seen the movie "A Christmas Prince"? It's another great Christmas romance film that's perfect for the holiday season. The chemistry between the lead actors, Ben Aldridge and Rose McIver, is undeniable. What do you think of that one?

User: oh that sounds perfect! i love emilia clarke so much, she was amazing as daenerys. last christmas sounds exactly like what im looking for - a recent christmas romance with someone i recognize. ill definitely watch that one tonight! thanks so much for the recommendation! [[TERMINATE CHAT]]